LESSON PLAN · SCRATCH (GR 6–8) ADVANCED: ALGORITHMS LAB

Lesson 9: Project: Secret Messenger

Module 3 | 50 minutes | Project: cryptography app | Grades 6–8 · ages 12–14

Objective: Plan, build and test an app that encrypts and decrypts messages with a chosen key, organised into procedures.

Success criteria

- My app encrypts and decrypts with a key.
- It uses one shift procedure.
- Round-trip tests pass.

Key vocabulary

- **encrypt / decrypt**: Scramble / unscramble a message.
- **inverse**: An operation that undoes another.
- **round-trip test**: Encrypt then decrypt to check you get the original.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)
- Printed project brief

Timed lesson plan (50 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Spy pairs (5 min): pass encrypted notes with cipher wheels and agreed keys.
5 min	**Watch & plan**	Play the project video; learners plan on the project brief.
25 min	**Build**	Build-along steps, then Round trip. Check in halfway: everyone has something working.
7 min	**Test & peer feedback**	Swap and test each other's work: two stars and a wish.
8 min	**Improve & submit**	Improve one thing; Project: Secret Messenger.

Unplugged warm-up

Spy pairs (5 min): pass encrypted notes with cipher wheels and agreed keys.

Teaching the idea

Modular design: a single parameterised shift procedure serves encryption and decryption (inverse key). Round-trip testing verifies inverse functions. Validation of menu and key.

Analogy: A lock and key: the same key locks and unlocks, turned the other way.

Live demo: Show that encrypting with key 3 then −3 returns the original.

Questions to ask

Ask	Listen for
How does one procedure do both jobs?	Opposite shift amounts.
What does a round-trip test prove?	Encrypt and decrypt are inverses.
How could you strengthen the cipher?	Different shift per letter (Vigenère).

Differentiation

Support
Provide the shift procedure; learners build the menu.

Stretch
A Vigenère cipher with a keyword (different shift per letter).

Common misconceptions

Misconception	How to fix it
Decryption needs separate code	Use the inverse shift.
Ciphers make messages totally safe	Caesar is easy to break.

Assessment

Project rubric (also on the project brief):

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Cipher	Incorrect	Encrypts only	Encrypts and decrypts with wrap-around	Handles spaces, capitals and punctuation
Procedures	None	Separate copies	One shift procedure for both	Clean, reusable design
Robustness	Breaks	Some checks	Validated menu and key	Friendly handling of all inputs
Testing	None	Ran once	Three round-trip tests	Systematic tests including edge letters
Security reflection	None	Vague	Explains weaknesses	Proposes and justifies an improvement

Exit ticket
What is the decrypt shift for key 7?

Homework
Exchange encrypted messages with a friend using your apps.

Answer key — Kitabash activities

1. Round trip (perform · ordering)

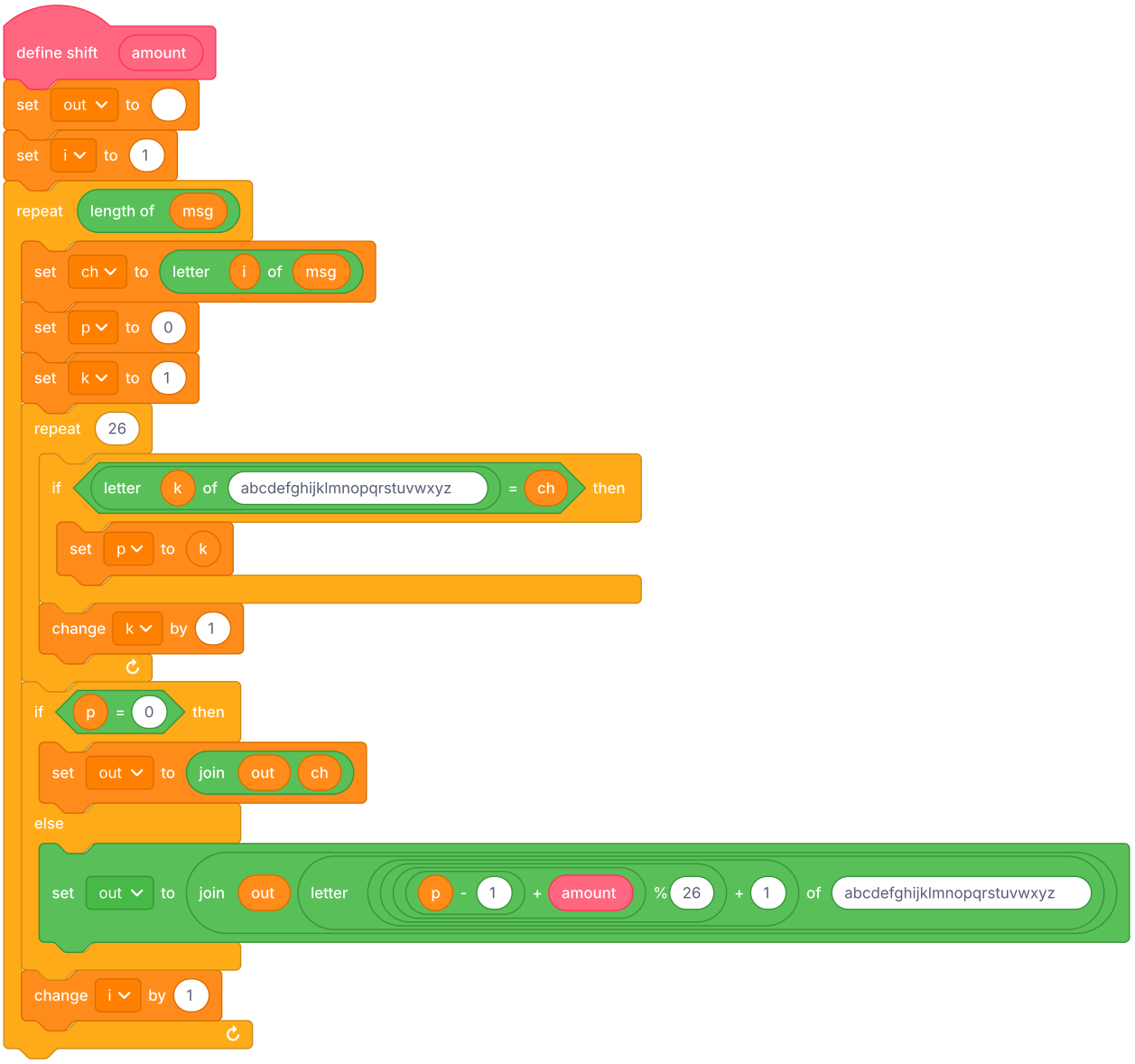
Order: Choose a message and a key → Encrypt the message → Copy the encrypted text → Decrypt it with the same key → Check you got the original message

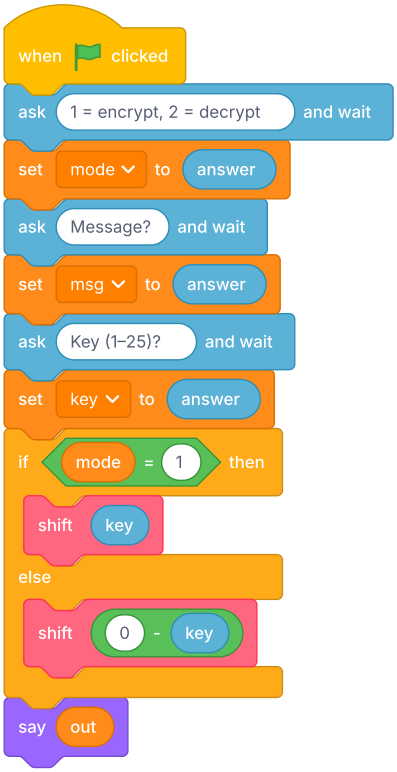
2. Project: Secret Messenger (probe · code lab)

Goals checked: Encrypts · Decrypts · Uses a procedure · Uses the key input

Example project:

Agent


define shift amount
set out to
set i to 1
repeat length of msg
set ch to letter i of msg
set p to 0
set k to 1
repeat 26
if letter k of abcdefghijklmnopqrstuvwxyz = ch then
set p to k
change k by 1
if p = 0 then
set out to join out ch
else
set out to join out letter p - 1 + amount % 26 + 1 of abcdefghijklmnopqrstuvwxyz
change i by 1


when clicked
ask 1 = encrypt, 2 = decrypt and wait
set mode to answer
ask Message? and wait
set msg to answer
ask Key (1–25)? and wait
set key to answer
if mode = 1 then
shift key
else
shift 0 - key
say out

Answer key — worksheet

1. **Key words:** encrypt / decrypt = Scramble / unscramble a message. · inverse = An operation that undoes another. · round-trip test = Encrypt then decrypt to check you get the original.
2. **Round trip:** 1 Choose a message and a key · 2 Encrypt the message · 3 Copy the encrypted text · 4 Decrypt it with the same key · 5 Check you got the original message
3. **My plan:** A clear plan that covers every requirement.
4. **Checklist:** All requirements met.
5. **Two stars and a wish:** Kind, specific feedback and an improvement.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.